GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Advanced Machine Learning Mini Project

Ensemblemethods(Bagging,RandomForest,AdaBoost,GradientBoosting)forhandwritten-digit recognition

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Subject	Machine Learning (METCE11504)	Semester	1 (M.Tech)
Activity	ALA 3	Marks	10

Deliverables: this concise project report and the executed notebook `ML_ALA3_Ensemble_Mini_Project_Notebook.ipynb`(complete code and outputs).

1. Problem Formulation

Task: classify handwritten digits (0–9) from small 8 × 8 grey-scale images (64 pixel features): a 10-class problem. **Data:** the *Optical Recognition of Handwritten Digits* data set (UCI; included in scikit-learn): 1,797 images,about180perclass,nomissingvalues.Low-resolutionimagesaretypicalofcheapsensorsinIoT devices (for example, reading digits on a meter), where a compact and accurate model is needed.

Advancedconcept:ensemblelearning. Asingledecisiontreeisunstable(asmallchangeofthetraining data changes the tree). Ensembles combine many trees:

Method	Idea	Main effect
Bagging	Trees trained on bootstrap samples; votes are combined	Reduces variance
Random Forest	Baggingplusarandomsubsetoffeaturesateachsplit (de-correlates the trees)	Reducesvariance further
AdaBoost	Trees trained in sequence; misclassified samples get larger weights	Reduces bias
Gradient Boosting	Each new tree fits the errors (gradient) of the current ensemble	Reducesbias;high accuracy

Objectives: (1) compare a single decision tree with the four ensembles; (2) measure bias and variance empirically; (3) study the effect of the number of trees and of tuning; (4) analyse performance, cost and errors; (5) propose future scope.

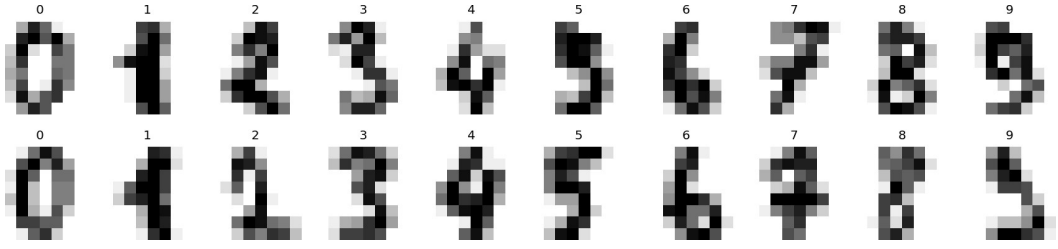


Figure 1: Sample images from the data set (true label above each image).

2. Model Implementation and Experimental Set-up

Item	Specification
Split	Stratified: 1,347 training and 450 test images. Tree models need no scaling.
Models	Decisiontree;Bagging(100trees);RandomForest(300trees);AdaBoost(200treesofdepth3, learning rate 0.5); Gradient Boosting (100 trees, depth 3, learning rate 0.1); RBF-SVM as a non-ensemble reference.
Evaluation	5-foldstratifiedcross-validationonthetrainingset;testaccuracyandmacro-F1;trainingtime, prediction time and model size (total tree nodes).
Further analyses	Accuracyversusnumberoftrees;out-of-bag(OOB)score;gridsearchfortheRandomForest; empirical bias–variance decomposition (20 re-trainings on random 80% subsets); confusion matrix and pixel importance.
Tools	Python, scikit-learn, NumPy, pandas, Matplotlib (Jupyter notebook).

3. Results and Performance Analysis

Comparison of models

Model	CVaccuracy (mean ± sd)	Test accuracy	Test macro-F1	Train time(s)	Predict(ms/image)	Tree nodes
Decision Tree (single)	84.9% ± 1.4	82.4%	0.823	0.02	0.001	265
Bagging (100 trees)	94.3% ± 0.4	92.4%	0.923	1.07	0.048	19,494
Random Forest (300 trees)	97.7% ± 1.1	96.9%	0.968	0.85	0.075	97,572
AdaBoost (200 stumps-depth3)	95.0% ± 0.5	95.1%	0.951	1.43	0.055	2,994
GradientBoosting(100 trees)	96.1% ± 1.3	95.8%	0.958	6.10	0.021	14,862
SVM(RBF)-non-ensemble reference	99.0% ± 0.4	99.1%	0.991	0.02	0.036	-

Timings are from one machine (Python) and are indicative only.

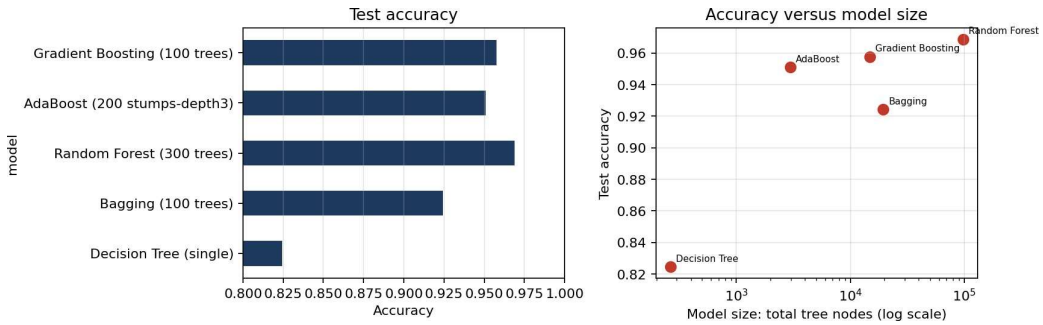


Figure 2: Test accuracy of the models and the trade-off between accuracy and model size.

A single decision tree reaches only 82.4%. All ensembles are clearly better: Bagging 92.4%, AdaBoost 95.1%, Gradient Boosting 95.8% and Random Forest 96.9%. **The RBF-SVM reference reached 99.1% and was better than every ensemble**, so ensembles are not automatically the best model family on a given problem.

Effect of the number of trees; out-of-bag score; tuning

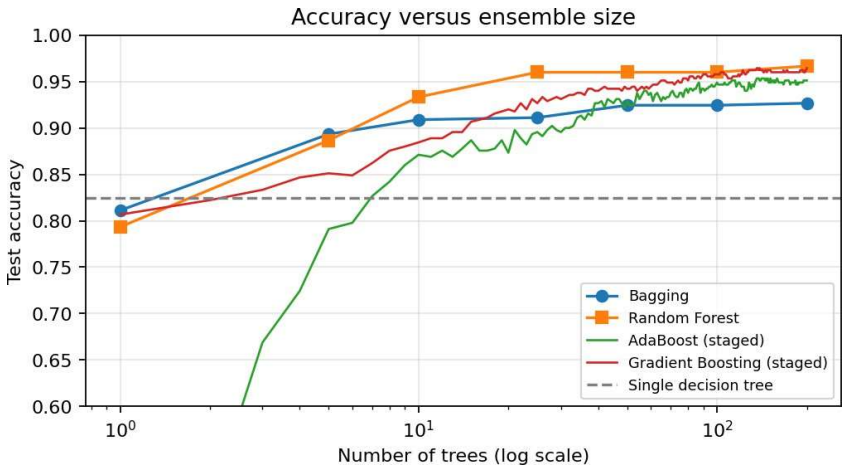


Figure 3: Test accuracy versus number of trees (boosting curves show the accuracy after each added tree).

- The Random Forest reaches 95% accuracy with about 25 trees and then flattens at 96–97%; Bagging improves slowly (92.7% with 200 trees), likely because its trees stay strongly correlated (Breiman, 2001).
- Theout-of-bagestimateoftheforest(97.6%)isclosetothetestaccuracy(96.9%):itisafreevalidation score.
- Grid search over max_features, min_samples_leaf and max_depth selected the default settings ({'max_depth': None, 'max_features': 'sqrt', 'min_samples_leaf': 1}); the tuned 200-tree forest scored 96.7% on the test set, one image different from the default, so tuning gave **no improvement**.

Bias–variance analysis

Model	Bias	Variance	Average error
Decision Tree	0.109	0.131	0.175
Bagging (25)	0.069	0.039	0.086
Random Forest (100)	0.033	0.010	0.035
AdaBoost (100)	0.049	0.033	0.065
Gradient Boosting (50)	0.056	0.033	0.065

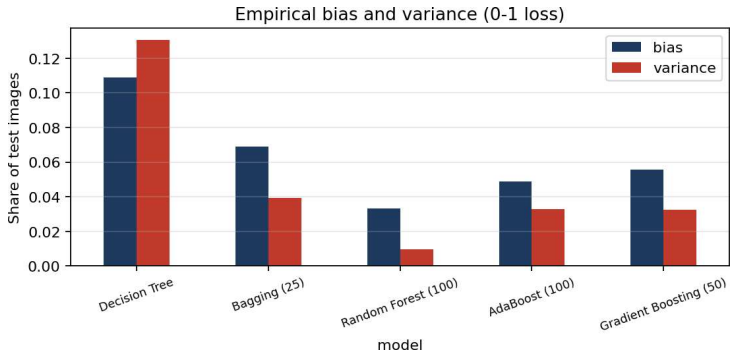


Figure4:Empiricalbiasandvariance(0-1loss,20re-trainingsonrandom80%subsetsofthetrainingdata;simplified decomposition).

The single tree has high variance (0.131): its prediction changes for about 13% of the test images when the training data change slightly. Bagging and the Random Forest cut the variance to 0.039 and 0.010, andthebiasalsofalls(tree0.109,RandomForest0.033,AdaBoost0.049,GradientBoosting0.056).The

Random Forest has the lowest average error among the methods compared.

Error analysis

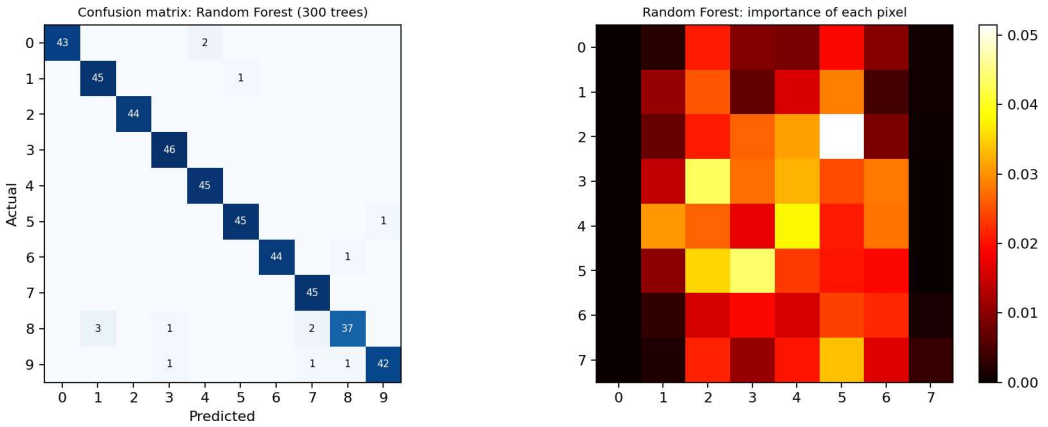


Figure 5: Confusion matrix of the best ensemble and pixel importance map of the Random Forest.

The best ensemble (Random Forest (300 trees)) made 14 errors on 450 test images. The hardest digitsare[8,9,1](F1[0.9,0.95,0.96]);themostfrequentconfusions(count,true→predicted)are3(8->1),2 (8->7),2(0->4),1(9->8),1(9->7).Pixelimportanceconcentratesintheinterioroftheimage;thefour corner pixels have importance of at most 0.0035 (mean 0.0156) because they are almost always blank.

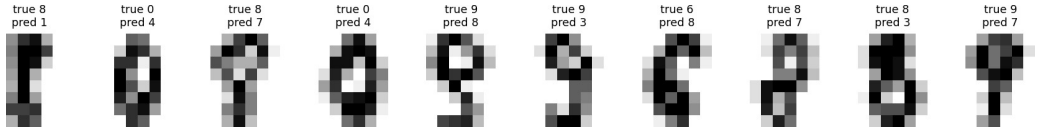


Figure 6: Misclassified test images (true and predicted labels).

4. Observations and Conclusions

- **Ensembles help a lot compared with one tree:** test accuracy rose from 82% to 92–97%, and variance fell by a factor of 3 to 13.
- **Bagging-type versus boosting-type:** bagging mainly reduces variance; boosting mainly reducesbias. Among the ensembles the Random Forest was the most accurate and easiest to use (no scaling, almost no tuning); Gradient Boosting and AdaBoost followed.
- **Cost matters:** the Random Forest has about 97,600 tree nodes against about 3,000 for AdaBoost and 265 for one tree, and Gradient Boosting needs the longest training (about 6 s). On small devices a moderate number of shallow trees may be preferable.
- **No universal winner:** the RBF-SVM was better than all ensembles on this small, smooth data set; model families must be compared on the problem at hand.
- **Limitations:** small, clean, low-resolution data; one train/test split plus cross-validation; no noise or distribution shift; bias–variance estimated from only 20 re-trainings; single-machine timings.

5. Future Scope

- **IoT / edge deployment:** prune the forest, limit depth, quantise thresholds, and measure latency and energy on a microcontroller-class device.
- **Deep learning:** compare with a small convolutional neural network and with XGBoost/LightGBM on larger, noisier images.

- **Robustnessandupdating:**testsensornoiseanddrift;useonlinelearningtoupdatethemodelonthe device; use semi-supervised or active learning to reduce labelling effort.
- **Trust:** use permutation importance or SHAP, calibrated probabilities and a reject option for uncertain images.

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